

Matthew Jörke

PhD Candidate, Computer Science

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EDUCATION

- 2020—present Stanford University
PhD Candidate, Computer Science
Advisors: James A. Landay, Emma Brunskill
- 2015—2019 University of California, Berkeley
BA, Computer Science; BA, Cognitive Science
with Highest Distinction in General Scholarship
with Honors in EECS (honors concentration in Cognitive Neuroscience)
- 2019 Hochschule für Gestaltung Schwäbisch Gmünd
Exchange Semester in Interaction Design

RESEARCH EXPERIENCE

- 2025 Apple
Research Intern
Advisors: Nick Foti, Guillermo Sapiro
- 2022 Microsoft Research
Research Intern
Advisors: Gonzalo Ramos, Jina Suh, Mary Czerwinski
- 2019—2020 School of Information, UC Berkeley
Research Assistant
Advisors: Jon Gillick, David Bamman
- 2018—2019 Hybrid Ecologies Lab, UC Berkeley
Research Assistant
Advisors: Cesar Torres, Eric Paulos
- 2017—2018 D'Esposito Lab, UC Berkeley
Research Assistant
Advisors: Justin Riddle, David Vogelsang, Mark D'Esposito
- 2016 Knight Lab, UC Berkeley
Undergraduate Research Apprenticeship Program
Advisor: Ignacio Saez

PUBLICATIONS

* indicates equal contribution

Peer-Reviewed Academic Publications

- CHI '26 Bloom: Designing for LLM-Augmented Behavior Change Interactions.
Matthew Jörke, Defne Genç*, Valentin Teutschbein*, Shardul Sapkota, Sarah Chung, Paul Schmiedmayer, Maria Ines Campero, Abby C. King, Emma Brunskill, James A. Landay. 2026. *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*.
🏆 Best Paper Award (Top 1%)
- CHI '25 GPTCoach: Towards LLM-Based Physical Activity Coaching.
Matthew Jörke, Shardul Sapkota, Lyndsea Warkenthien, Niklas Vainio, Paul Schmiedmayer, Emma Brunskill, James A. Landay. 2025. *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*.
- AAAI '25 Cost-Aware Near-Optimal Policy Learning.
Joy He-Yueya, Jonathan Lee, Matthew Jörke, Emma Brunskill. 2025. *Proceedings of the AAAI Conference on Artificial Intelligence*.
- CHIL '24 Adaptive Interventions with User-Defined Goals for Health Behavior Change.
Aishwarya Mandyam*, Matthew Jörke*, William Denton, Barbara E. Engelhardt, Emma Brunskill. 2024. *Proceedings of Machine Learning Research: Conference on Health, Inference, and Learning*.
- CSCW '24 “They Make Us Old Before We're Old”: Designing Ethical Health Technology with and for Older Adults.
Jianna So, Samantha Estrada, Matthew Jörke, Eva Bianchi, Maria Wang, Nava Haghighi, Kristen L. Fessele, James A. Landay, Andrea Cuadra. 2024. *Proceedings of the ACM on Human-Computer Interaction, Volume 8, Issue CSCW2*.
🏆 Best Paper Honorable Mention Award (Top 3%)
- CSCW '24 Improving Work-Nonwork Balance with Data-Driven Implementation Intention and Mental Contrasting.
Yasaman S. Sefidgar, Matthew Jörke, Jina Suh, Koustuv Saha, Shamsi Iqbal, Gonzalo Ramos, Mary Czerwinski. 2024. *Proceedings of the ACM on Human-Computer Interaction, Volume 8, Issue CSCW2*.
- DIS '23 A Workshop-Based Method for Navigating Value Tensions in Collectively Speculated Worlds.
Nava Haghighi*, Matthew Jörke*, Yousif Mohsen, Andrea Cuadra, James A. Landay. 2023. *Proceedings of the 2023 ACM Designing Interactive Systems Conference*.
- IUI '23 Pearl: A Technology Probe for Machine-Assisted Reflection on Personal Data.

Matthew Jörke, Yasaman S. Sefidgar, Talie Massachi, Jina Suh, Gonzalo Ramos. 2023. *Proceedings of the 28th International Conference on Intelligent User Interfaces*.

CSCW '23 Explanations Can Reduce Overreliance on AI Systems During Decision-Making.
Helena Vasconcelos, Matthew Jörke, Madeleine Grunde-McLaughlin, Tobias Gerstenberg, Michael S. Bernstein, Ranjay Krishna. 2023. *Proceedings of the ACM on Human-Computer Interaction, Volume 7, Issue CSCW1*.
🏆 Best Paper Honorable Mention Award (Top 3%)

Findings of EMNLP '20 Attending to Long-Distance Document Context for Sequence Labeling.
Matthew Jörke, Jon Gillick, Matthew Sims, David Bamman. 2020. *Findings of the Association for Computational Linguistics: EMNLP 2020*.

C&C '19 Hybrid Microgenetic Analysis: Using Activity Codebooks to Identify and Characterize Creative Process.
Cesar Torres, Matthew Jörke, Emily Hill, Eric Paulos. 2019. *Proceedings of the 2019 Conference on Creativity and Cognition*.

Nonarchival Papers, Preprints, and Book Chapters

Preprint Reflections and New Directions for Human-Centered Large Language Models.
Caleb Ziems, Dora Zhao, Rose E. Wang, Matthew Jörke, et al., 2026. *arXiv*.

Preprint LLMs Are Not (Consistently) Bayesian: Quantifying Internal (In)consistencies of LLMs' Probabilistic Beliefs.
Chacha Chen*, Matthew Jörke*, Adam Goliński, Masha Fedzechkina, Guillermo Sapiro, Sinead Williamson, Nicholas Foti. 2026. *arXiv*.

Preprint "What Are You Really Trying to Do?": Co-Creating Life Goals from Everyday Computer Use.
Shardul Sapkota, Matthew Jörke*, Zane Sabbagh*, Omar Shaikh, Grace Wang, James A. Landay. 2026. *arXiv*.

Preprint Generative Experiences for Digital Mental Health Interventions: Evidence from a Randomized Study.
Ananya Bhattacharjee, Michael Liut, Matthew Jörke, Diyi Yang, Emma Brunskill. 2026. *arXiv*.

CHI '23 Extended Abstracts Lessons Learned for Data-Driven Implementation Intentions with Mental Contrasting.
Yasaman S. Sefidgar, Matthew Jörke, Jina Suh, Koustuv Saha, Shamsi Iqbal, Gonzalo Ramos, Mary Czerwinski. 2023. *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*.

Design Thinking Research	User Perceptions of Privacy Interfaces in the Workplace. Michelle S. Lam, <u>Matthew Jörke</u> , Jennifer King, Nava Haghighi, James A. Landay. 2023. <i>Design Thinking Research: Innovation—Insight—Then and Now</i> .
ICML ‘22 Workshop	Simple Regret Minimization for Contextual Bandits Using Bayesian Optimal Experimental Design. <u>Matthew Jörke</u> , Jonathan Lee, Emma Brunskill. <i>ICML Workshop on Adaptive Experimental Design and Active Learning in the Real World</i> .
CHI ‘22 Workshop	When Do XAI Methods Work? A Cost-Benefit Approach to Human-AI Collaboration. Helena Vasconcelos, <u>Matthew Jörke</u> , Madeleine Grunde-McLaughlin, Tobias Gerstenberg, Michael S. Bernstein, Ranjay Krishna. 2022. <i>CHI Workshop on Trust and Reliance in AI-Human Teams</i> .

RESEARCH MENTORSHIP

* indicates co-authorship on publications

2025—present	William Fang (MS)
2024—2025	Defne Genç* (MS)
2024—2025	Sarah Chung* (MS)
2024—2025	Valentin Teutschbein* (BS)
2024	Andrew Sung (BS)
2024	Bryant Jimenez (BS)
2023—2024	Lyndsea Warkenthien* (MS)
2023—2024	William Denton* (BS/MS)
2023	Anthony Xie (BS)
2023—2024	Niklas Vainio* (BS/MS)
2023	Niall Kehoe (BS)
2023—2024	Romuald Thomas (BS/MS)
2023—2024	Taylore Givens (BS)
2022	Rachel Ka Hei Wu (BS)
2022	Nahum Maru (BS)
2021—2022	Roland Martinez Luna (BS)
2021—2023	Yousif Mohsen* (BS)
2021	Luke Mann (BS)
2021	Rishi Sreekanth (BS)

GRANTS, FELLOWSHIPS, AND AWARDS

2024—present	Hasso Plattner Institute & Stanford Human-Centered AI Fellowship
2024	Best Paper Honorable Mention, ACM CSCW (top 3% of papers)
2023—2024	Stanford McCoy Center for Ethics in Society Graduate Student Fellowship
2023—2024	Stanford HAI Seed Grant
2023	Best Paper Honorable Mention, ACM CSCW (top 3% of papers)
2021—2022	Hasso Plattner Design Thinking Research Program Research Grant
2021—2022	Stanford Ethics, Society, and Technology Hub Seed Grant

2017—2019	UC Berkeley EECS Honors Program
2015	Edward Frank Kraft Award

TEACHING

2026	Head Course Assistant Stanford CS 234: Reinforcement Learning <i>Instructor:</i> Emma Brunskill
2023	Project Mentor CS 342/MED 253: Building for Digital Health <i>Instructors:</i> Oliver Aalami, Paul Schmiedmayer, Vishnu Raavi
2023	Head Course Assistant Stanford CS 147: Introduction to Human-Computer Interaction <i>Instructor:</i> James Landay
2020	Teaching Assistant UC Berkeley Info 159: Natural Language Processing <i>Instructor:</i> David Bamman
2019	Workshop Instructor HfG Schwäbisch Gmünd, Internet of Things Department Invited to teach a three-part workshop on AI for design students

REVIEWING

2023—2025	CHI
2025	Nature
2025	IUI
2025	CSCW
2024	IMWUT
2024	IEEE Pervasive
2022	UIST

SERVICE

2025—present	Recovery Companion Foundation, Board Member
2022—2024, 2025—present	Stanford HCI Lunch (Co-)Organizer
2020, 2021, 2025	Stanford Student Application Support (Reviewer)
2023	IUI Conference Student Volunteer
2020—2022, 2025	Stanford CS Undergraduate Mentoring Program
2020—2022	Stanford Computing & Society Co-Organizer